Paul B. Sharp

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Academic Appointments

Present — **Bar-Ilan University**, Assistant Professor, Psychology Department, Gonda Multidisciplinary Brain Research Center

2023 – 2024 — **Yale University**, Postdoctoral Research Associate

2021 – 2023 — **Hebrew University**, Fulbright Postdoctoral Associate

2019 – 2021 — **Max Planck UCL Center for Computational Psychiatry**, Postdoctoral Associate

Education

2016 – 2019 — **University of North Carolina at Chapel Hill**, PhD, Psychology

2013 – 2016 — **University of Illinois at Urbana-Champaign**, MA, Psychology (PhD Track)

2010 – 2013 — **Temple University**, BA, Psychology, Political Science, *Magna Cum Laude*

Awards and Funding

2025 – 2030 — Bereshit Grant, Israeli Science Foundation, Value: $810,000

2024 – 2027 — Azrieli Early Career Faculty Fellowship, Azrieli Foundation, Value: $202,000.

2025 — Top-10 Publication Prize ("Publish & Take"), Bar-Ilan University, for a publication in a top-decile journal (JCR), Value: 10,000 NIS.

2021 – 2023 — Fulbright Postdoctoral Fellowship, Fulbright Foundation, Value: $95,000.

2022 — eLife Early-Career Reviewer.

2008 — American Mathematics Competition Honor Roll, MAA.

Publications

2026 — Anxiety is associated with biases in task generalization. Schiff H, Sharp PB. *Topics in Cognitive Science.*

2025 — Anxiety involves altered planning. Sharp PB. *Trends in Cognitive Sciences.*

2025 — Garvert MM, McFadyen J, Linke S, McCloud T, Meyer SS, Sobanska S, Sharp PB, Long A, Huys QJM, Ahmadi M. Safety and efficacy of a modular digital psychotherapy for social anxiety: A randomized controlled trial. *JMIR.*

2024 — Humans adaptively deploy forward and backward prediction. Sharp PB, Eldar E. *Nature Human Behaviour.*

2023 Hierarchical inference as a source of human biases. Sharp PB, Fradkin I, Eldar E.
Cognitive, Affective, and Behavioral Neuroscience.

2022 Humans perseverate on punishment avoidance goals in multigoal reinforcement learning. Sharp PB, Russek E, Huys Q, Dolan RJ, Eldar E.
eLife.

2022 Training diversity promotes absolute-value-guided choice. Solomyak L, Sharp PB, Eldar E.
PLOS Computational Biology.

2021 Disrupted state transition learning as a computational marker of compulsivity. Sharp PB, Dolan RJ, Eldar E.
Psychological Medicine.

2021 Cognitive control deployment is flexibly modulated by social value in early adolescence. Sharp PB, Do KT, Lindquist KA, Prinstein MJ, Telzer EH.
Developmental Science.

2020 Towards formal models of psychopathological traits that explain symptom trajectories. Sharp PB, Miller GA, Dolan RJ, Eldar E.
BMC Medicine.

2020 Modernizing conceptions of valuation and cognitive control deployment in adolescent risk taking. Do KT*, Sharp PB*, Telzer EH.
Current Directions in Psychological Science.

2019 Computational models of anxiety: Nascent efforts and future directions. Sharp PB, Eldar E.
Current Directions in Psychological Science.

2019 Mechanistic science: A new approach to comprehensive psychopathology research that relates psychological and biological phenomena. Thomas JG, Sharp PB.
Clinical Psychological Science.

2019 Reduction and autonomy in psychology and neuroscience: A call for pragmatism. Sharp PB, Miller GA.
Journal of Theoretical and Philosophical Psychology.

2018 Mindfulness training induces structural connectome changes in insula networks. Sharp PB, Sutton BP, Paul EJ, et al.
Scientific Reports.

2017 Structural connectomics of anxious arousal in early adolescence: Translating clinical and ethological findings. Sharp PB, Telzer EH.
NeuroImage: Clinical.

2017 Selective neural sensitivity to familial threat in adolescents with weak family bonds. Sharp PB, Heller W, Telzer EH.
Social Neuroscience.

2015 Transdiagnostic dimensions of anxiety: Neural mechanisms, executive functions, and new directions. Sharp PB, Miller GA, Heller W.
International Journal of Psychophysiology.

Accepted in Principle Bedder RL, Hitchcock PF, Sharp PB. Unravelling Repetitive Negative Thinking With Reinforcement Learning.
Clinical Psychological Science.

In Revision Sharp PB, Eldar E. Humans adaptively delay planning using cognitive maps.
Psychological Science (post first revision).

Under Review Sharp PB. A World-Model Theory of Trait Anxiety.
Psychological Review.

Book Chapters Sharp PB. Computational Models of Anxiety. Invited chapter, accepted, in Y.-K. Kim (Ed.), *Anxiety Disorders, 2nd ed.: Current Challenges and Future Perspectives.*

In Preparation Yadid N, Linkovski O, Sharp PB. Inter-trial intervals induce learning biases.
Three replicated, pre-registered studies (total $N = 700$); Best Poster Award, Computational Psychiatry Conference, Yale University, 2026.

Pizanti A, Sharp PB. Worry is associated with excessive model-based revaluation.
Two replicated, pre-registered studies (total $N = 300$); Poster Award Lecture, Computational Psychiatry Conference, Yale University, 2026.

Sharp PB, Feng GW, Rutledge RB. Early outcomes shift Pavlovian biases.

Fradkin I, Sharp PB. Offline learning biases in psychopathology.

Conference Symposia

July 2025 Chair. Cognitive map biases across psychopathology. Computational Psychiatry Conference, Tübingen, Germany.

June 2022 Co-Chair. Repetitive Negative Thinking and Simulation in Natural and Artificial Cognition. The 5th Multidisciplinary Conference on Reinforcement Learning and Decision Making, Brown University.

Fall 2017 Co-Chair. Constructing riskier tests of theory, linking psychological and biological phenomena, and introducing a new mechanistic philosophy of science. Society for Psychophysiological Research, Vienna, Austria.

Invited Talks

February 2027 **Invited colloquium (scheduled; paid).**
Agnes Scott College, Psychology Department, Atlanta, GA.

July 2025 **Task inference in planning specifies anxious generalization failures**
University of Würzburg, Psychology Department (paid; funded by a DFG training grant).

April 2025 **Invited research visit and talk (paid).**
Zurich Center for Neuroeconomics (host: Todd Hare), University of Zurich.

February 2025 **Meta-control and Planning**
Emory University, Psychology Department.

January 2025 **Why plan forward or backward?**
Rick Adams Lab, UCL.

July 2024 **Over-planning computations in worry.**
Computational Psychiatry Conference, Minneapolis, MN.

June 2022 **Using a science of planning to understand the dynamics of worry.**
The 5th Multidisciplinary Conference on Reinforcement Learning and Decision Making.

August 2022 **Understanding worry using planning algorithms.**
Laboratory of Neural Computation and Cognition (Michael J. Frank Lab), Brown University, RI.

February 2022 **Efficient planning algorithms humans use to pursue multiple goals.**
Israeli Society for Cognitive Psychology.

September 2021 **Computations of Chronic Worry: Dysfunction in Multi-goal Learning and Planning Efficiency.**
Society for Research in Psychopathology.

March 2021 **Neuroscience only leads to understanding with computation: A brief history of efforts to understand human anxiety.**
Leelab, Virginia Tech, VA.

December 2019 **How task-irrelevant emotion affects model-building: Implications for basic learning processes and psychopathology.**
Tel Aviv University, Bar-Haim Lab.

October 2019 **Computations in human anxiety: New developments in a nascent field.**
National Institute of Mental Health, Pine Lab, Bethesda, MD.

Teaching

Courses developed at Bar-Ilan University:
Computational Psychiatry Modelling (MA)
Introduction to Computational Psychiatry (BA)
New Frontiers in the Science of Anxiety (MA & BA)
Introduction to Psychology (BA)
Introduction to Statistics (BA)
Practicum on AI and Mental Health (BA)

Mentorship

Fall 2026 — **Daniel Toledano**, incoming Postdoctoral Researcher, Bar-Ilan University.

Fall 2026 — **Noam Gil-Adi** (MA, Ben-Gurion University), incoming Post-baccalaureate Research Assistant, Bar-Ilan University.

Fall 2026 — **Diana Vrubel** (MSc, Higher School of Economics), incoming PhD Student, Gonda Neuroscience PhD Track, Bar-Ilan University.

Present — **Adi Pizanti**, MA Student, Bar-Ilan University.
Poster Award Lecture, Computational Psychiatry Conference, Yale University, 2026.

2024 – Present — **Noa Yadid**, Bar-Ilan University.
Best Poster Award, Computational Psychiatry Conference, Yale University, 2026.

Present — **Dafna Yavetz**, Graduate Student, Bar-Ilan University.
Selected, with full funding, for NeuroBridges 2026 summer school in computational neuroscience (faculty incl. L. Abbott, R. Shadmehr, M. Pessiglione), Cluny, France.

2024 – 2025 — **Marine Hainguerlot**, Postdoctoral Researcher, Bar-Ilan University.

Mentored Student Conference Posters

2026 — Pizanti A, Sharp PB. Worry is associated with excessive model-based revaluation.
Computational Psychiatry Conference, Yale University. **Poster Award Lecture.**

2026 — Yadid N, Linkovski O, Sharp PB. Inter-trial intervals assign time costs to actions during learning.
Computational Psychiatry Conference, Yale University. **Best Poster Award.**

2026 — Yavetz D, Sharp PB. How model-based reinforcement learning degrades in aging.
Computational Psychiatry Conference, Yale University.

2026 — Yavetz D, Sharp PB. Metacontrol Efficiency in Aging: A Pilot Study.
Israeli Society of Biological Psychiatry, Annual Meeting.

Faculty Service

2024, 2025 — Chair, PhD Symposium, Psychology Department, Bar-Ilan University.

2025 — Chair, Psychology Department Conference, Bar-Ilan University.

Grant Reviewing and Faculty Mentorship

2024 – Present — Formal reviewer, Israel Science Foundation (two proposals).

2024 – Present — Formal reviewer, German Research Foundation (DFG) grant proposal.

2024 – Present — Formal reviewer, Huo Family Foundation (London) grant proposal.

2025 — Faculty mentorship: formally reviewed and mentored Dr. Moshe Glickman's Israel Science Foundation proposal (funded).

Professional Memberships

2023 – Present Reinforcement Learning and Decision Making.

2023 – Present Computational Psychiatry.

Ad Hoc Reviewer

Neuron, Trends in Cognitive Sciences, Biological Psychiatry, eLife, Nature Communications, PLOS Computational Biology, Current Opinion in Behavioral Sciences, Journal of Neuroscience, Neuroscience and Biobehavioral Reviews, Clinical Psychology Review, and several others.